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Review Article

Reassessing academic integrity in the age of AI: A systematic literature review on AI and academic integrity

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ABSTRACT

Academic integrity is a key factor in the quality of education that represents honesty, trust, and ethical conduct. In today's rapidly changing educational landscape, artificial intelligence (AI) poses significant challenges to the educational ecosystem's ability to maintain academic integrity. It also affects the qualifications offered by institutes. Although AI supports students in completing academic tasks, it usually risks violating the basic rules of academic integrity. In addition, AI can be used to detect academic misconduct. This study aims to critically examine the impact of AI on academic integrity through a systematic literature review. The research question was developed using the PICO framework, and the articles considered in this study were collected from Scopus, PubMed, DOAJ, and Base. Of 1443 articles, 25 were selected based on the PRISMA framework. We used the Cochrane risk of bias tool (ROBINS-1) to analyse the risk of bias in the selected studies. The discussion section was developed based on PICO frameworks – population of the study, intervention with AI tools, comparison of modern and traditional methods, and outcome of AI use for academic activities – to answer the research question. This research contributes to the ongoing dialogue about AI and academic integrity by emphasising the importance of a balanced approach to using the benefits of AI in education while upholding ethical standards. This study concludes by emphasising the importance of creating a culture of academic integrity to ensure the ethical use of AI for educational purposes.

1. Introduction

Academic integrity is a critical component of education in today's rapidly changing academic landscape. Academic integrity must be maintained because it represents the value of the qualifications offered by an institute; the honesty, trust, and ethical conduct of students; and academic institutes' management of these factors. However, as artificial intelligence (AI) develops, traditional pedagogical practices may come with complex challenges and opportunities.

AI supports students in various aspects of academia, including data analysis, algorithmic decision-making, and writing. Because AI provides endless opportunities in educational settings, its integration into education has created complex issues related to academic integrity in traditional education. Meanwhile, academic institutions can reduce the risk of academic misconduct by using sophisticated algorithms to detect plagiarism, contract cheating, and data manipulation.

1.1. What Is academic Integrity?

Hagège (2023) mentioned that there is no proper definition of *academic integrity*.

However, some idea of its meaning is required for the present discussion. Simply put, it can be thought of as follows: one has to be honest in one's work, acknowledge others' work properly, and give credit where one has used other people's ideas or data (Campbell & Waddington, 2024). Academic integrity includes ethical standards and behaviour in academia. It consists of integrity, courtesy towards intellectual property, moral and ethical standards, and professional behaviour norms (Șercan & Voicu, 2022). It also indicates commitment, honesty, and moral behaviour in academic work done by both students and academics (Sbaffi & Zhao, 2022).

Furthermore, academic integrity is the collective activity of students and teachers; it is not possible to achieve academic integrity individually. High academic integrity can be achieved by examining factors such as cultural practices, agency, personality traits, learning ecosystems, and structure (Mathrani et al., 2021). The European Network for Academic

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Integrity (ENAI) developed a glossary for academic integrity that consists of 212 words. A broader explanation of academic integrity includes "compliance with ethical and professional principles, standards, practices and consistent system of values that serves as guidance for making decisions and taking actions in education, research, and scholarship" (Glossary – ENAI).

1.2. Academic Integrity from ancient times to the digital age

Plagiarism is the primary component of academic integrity. In the early stages, plagiarism was frowned upon in education. Therefore, it is worth examining where plagiarism began and how it is moving forward. Moss (2005) mentioned that Shakespeare stole historical plots from Holinshed. Furthermore, Roman poet Martial, who lived from approximately 40 AD to 104 AD, identified that his poetry was being copied and recited by other poets without his knowledge. Even though the word *plagiarism* was not introduced when Martial lived, similar acts were being committed by poets, and the word *plagiarism* was first entered in Samuel Johnson's dictionary in 1755 (Bailey, 2011). Various terms fall under the umbrella of academic integrity today, the most common of which are *contact cheating*, *ghostwriting*, and *AI writing*.

With the advancement of technology, communication has become fast and can connect anyone to others from anywhere in the world. This situation not only improves the infrastructure for student learning but also fosters a culture that may lead to breaches in academic integrity. On the internet, students can easily find contract writers and assignment helpers. There are also many AI writing software programmes available on the internet that write assignments. The latest such tool is ChatGPT, which was developed based on a generative pre-trained transformer method (M. Mijwil et al., 2023). Naidu and Sevnarayan (2023) mentioned that the use of ChatGPT in education has increased. Because ChatGPT can provide a personalised learning experience for students (Kasneci et al., 2023), its use is expected to expand. The problem is that even if students intend to use this technology ethically, academic integrity can be put at risk if students misuse it (Currie, 2023).

1.3. Artificial Intelligence and academic Integrity

The generative pre-trained transformer (GPT) model allows authors to write content in a second. This creates new opportunities for writing tasks (Biermann, 2022).

The resulting content fully machine-generated; the only human involvement is to generate a prompt. The language model does not have the understanding, critical thinking, or personal insights needed to produce authentic scholarly work. Consequently, this raises concerns about academic integrity because the content does not represent the author's ideas. If students use ChatGPT to write assignments, the originality of the assignments may decline (Liu et al., 2023). Moreover, when AI capabilities are used for academic cheating, the qualifications offered by the institute may not reflect the expected levels (Currie et al., 2023). Therefore, general performance is expected to reduce when students work on practical scenarios. However, if AI is used properly in educational contexts, it can enhance academic integrity and quality at all levels. For example, Grammarly, Wordtune, QuillBot, and similar software are helping students with language matters (Yeo, 2023) while offering better support to protect the academic integrity of institutes.

1.4. Purpose of the present study

While several researchers are addressing the issues of AI and academic integrity, the comprehensive studies exploring both the positive and negative effects of AI in education are still lacking. (Eaton, 2023; Chávez et al., 2023).

Eaton et al. (2023) discussed the importance of the use of AI in assessments, but often fail to discuss ethical AI use across diverse educational settings. Furthermore, educational institutions lack proper policies for acknowledging AI use in academic activities. Therefore, it is critical to maintaining the academic integrity(Foltynek et al., 2023; Gulumbe et al., 2024). Also, Akinwalere and Ivanov (2022) highlighted the AI in reshaping the educational landscape, but gaps still remain in understanding how to ethically manage these technologies in educational settings. While several studies discussed the benefits of AI in educational settings and automating administrative tasks, the ethical implications and potential for academic misconduct remain underexplored. Moreover, gaps are visible in the scant literature published on AI and academic integrity (Rodrigues et al., 2024).

Since AI continues to influence educational ecosystems, there is a need to critically examine the influence of AI on academic integrity. This need was pursued by conducting a systematic literature review to observe AI's application in higher education and its impact on academic integrity.

The purpose of the study is to critically examine the influence of artificial intelligence on academic integrity within the educational ecosystem. The review provides a comprehensive understanding of the role of AI in academia, highlighting its benefits and ethical challenges, and offers insights for a balanced approach to its integration in educational settings. We believe the present analysis and discussion of the relevant literature contribute to the ongoing dialogue on the ethical application of AI in academia.

2. Methods

The systematic reviews were protocol-driven and needed to be rigorously processed at each stage (Nightingale, 2009). Systematic reviews and meta-analyses (PRISMA) 2020 guidelines were used (Fig. 1). The PICO framework was used to develop the background of the research question (Richardson et al., 1995).

RQ: What is the role of AI in influencing academic integrity, and how can educational institutions ensure ethical AI usage?

Sub-research questions were developed according to the PICO framework (Table 1) to obtain a deeper understanding of the research issue (Table 2).

2.1. Search strategy

We conducted a literature search across four databases (Table 3). The search string was "artificial intelligence" OR "AI" AND "academic integrity" OR "educational integrity." The Boolean operators were "AND" and "OR". In total, N = 1443 records were found (n = 62 from PubMed, n = 1136 from DOAJ, n = 235 from Scopus, and n = 7 from Base, n = 3 expert recommendations). PRISMA guidelines were used to decide which papers to include and exclude to ensure the quality of the literature collection. Nested Knowledge's semi-automated software handled the include, exclude, and duplicate removal process (Nested Knowledge, 2024).

2.2. Inclusion and exclusion criteria

The database search query was run for the entire period of the database and was not filtered by publication year. Consequently, we collected all publications related to this query. The three steps included in the selection process were (1) duplicate identification, (2) screening, and (3) inclusion or exclusion. Thirty-two duplicate records (N = 32) were removed from the databases by using nested knowledge semi-automated software (Table 3).

The screening process included 1408 articles after duplicates were removed. The titles, keywords, abstracts, and eligibility of the PICO framework were thoroughly reviewed in the first screening phase, and some articles were excluded. During the second screening phase, we read the full text of each article that passed the first phase. During the second phase, articles were excluded for the following reasons: case reports (n = 17), conference abstracts (n = 6), editorial or opinion pieces

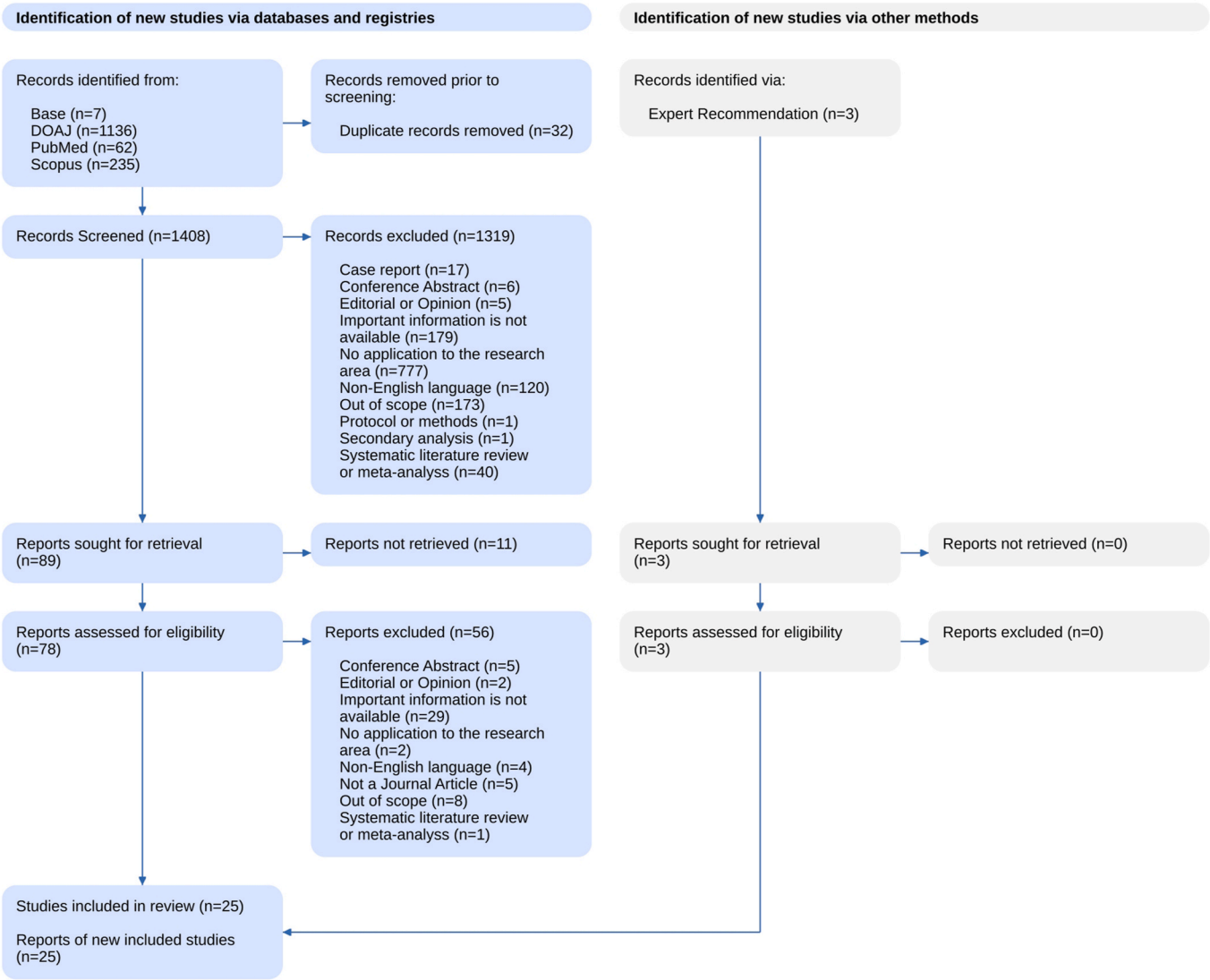

Fig. 1. The selected literature screening process using the PRISMA 2020 flow diagram for new systematic reviews.

Table 1
PICO framework.

Components of PICO	Components of Study Designs
Population	The population consists of students who use AI for their academic activities.
Intervention	The integration of AI tools and technologies such as ChatGPT and Google Bard.
Comparison	Traditional methods without AI integration.
Outcome	Identify the prevention of plagiarism, the detection of cheating using AI, and ethical considerations in AI usage for academic activities.

Table 2
Sub-research questions.

Components of PICO	Sub-Questions
Population	What characteristics do the participants have?
Intervention	What specific AI tools and technologies are being implemented in academic activities?
Comparison	What are the differences in academic integrity outcomes between assessments using AI and traditional methods?
Outcome	What ethical considerations emerge from the integration of AI in academic assessments, and how are they managed or mitigated?

(n = 5), important information is not available (n = 179), no application to the research area (n = 777), not in English (n = 120), beyond the scope of this study (n = 173), protocol or methods (n = 1), secondary analysis (n = 1), and systematic literature review and meta-analysis (n = 40). Eighty-nine reports were included in the retrieval process, and 11 were not retrieved. Therefore, n = 78 articles were included in the final pool. Also, n = 3 articles were included based on expert recommendations. After careful consideration of these articles, n = 25 were included in the final analysis (Fig. 1).

3. Results

The keyword occurrence map created by Vosviewer software (Van Eck & Waltman, 2010) (Fig. 2) indicates a network analysis of the collected literature. We arranged the collected data from the databases in a CSV file format compatible with VOSviewer. The data file included the title, abstract, keywords, authors, and year of publication. We removed duplicate entries before importing the file to VOSviewer. We used a minimum number of occurrences of 1, a total number of items of 84, and a threshold level of 84 for network visualisation. This helps in

Table 3
Database search summary.

Search	Database/Other Methods	Query	Extraction Method	Results	Duplicate	Excluded	Included
1	PubMed	"artificial intelligence" OR "AI" AND "academic integrity" OR "educational integrity"	Automated -API	62	0	58	4
2	DOAJ	"artificial intelligence" OR "AI" AND "academic integrity" OR "educational integrity"	Automated -API	1136	0	1134	2
3	Scopus	"artificial intelligence" OR "AI" AND "academic integrity" OR "educational integrity"	Manual Search	235	30	189	16
4	Base	"artificial intelligence" OR "AI" AND "academic integrity" OR "educational integrity"	Manual Search	7	2	5	0
5	Expert Recommendation		Manual Search	3	0	0	3
TOTAL				1443	32	1386	25

focusing on the most relevant keywords and reducing clutter.

The size of each node shows the frequency with which the corresponding keyword appears in the literature, while the thickness of each line shows the strength of the connection between the two keywords connected by that line. The colour bar indicates the publication duration; publication on this topic is indicated by yellow nodes.

Keywords in the selected literature are used to create a network visualisation map. The keyword occurrence map (Fig. 2) includes thin line connecting AI and academic integrity, indicating that a limited number of research covering both of these topics has been conducted. Therefore, the research gap exists between AI and academic integrity and a need to conduct a systematic literature review on these topics. Addressing this research gap may help explain how AI can be used in educational settings while maintaining academic integrity.

There are four clusters in this network (Table 4). Cluster 1 has four items, Clusters 2 and 3 include three items each, and Cluster 4 contains one item. *Academic integrity*, *academic misconduct*, and *AI* are significant words which included in Cluster 1. Claster 2 is basically focused on writing methods. The most important keyword in Cluster 3 is *quality assurance*, and in Cluster 4, the main keyword is *plagiarism*. These clusters represent the main research areas in the selected research papers. The fact that academic integrity and AI are both in Cluster 1 indicates a research gap in this cluster.

Table 4
Clusters of keywords in the network visualisation map Fig. 2.

Cluster	No. of Items	Items
1	4	Academic integrity, academic misconduct, AI, student character
2	3	Academic writing, generative AI, large language model
3	3	ChatGPT. higher education, quality assurance
4	1	Plagiarism

The most-cited paper is "Chatting and cheating: Ensuring academic

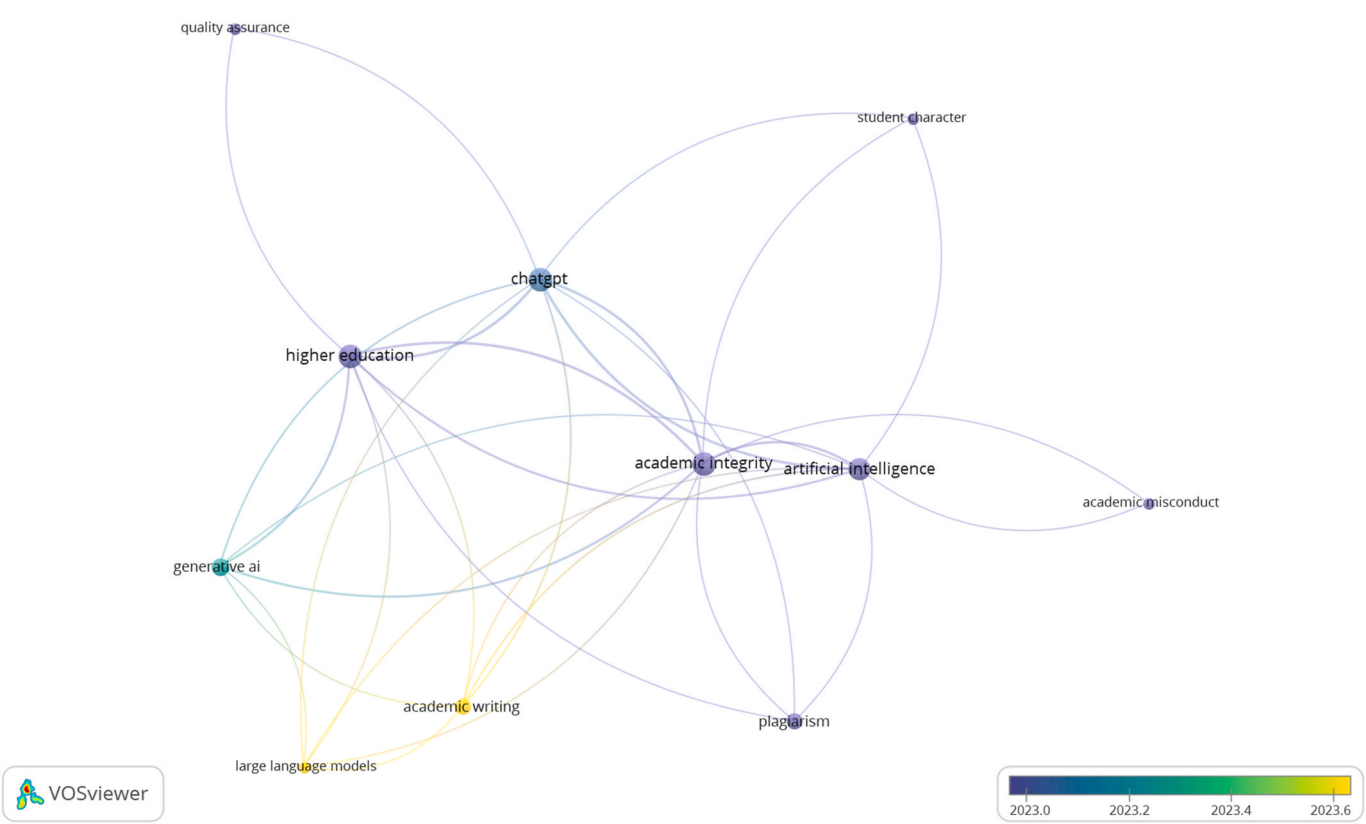

Fig. 2. Keyword co-occurrence network visualisation map
Minimum number of occurrences: 1, total number of items: 84, threshold level: 84, number of items: 11, number of clusters: 4.

integrity in the era of ChatGPT", with 755 citations. The second most-cited study is "ChatGPT in higher education: Considerations for academic integrity and student learning", with 221 citations (Table 5). "Leadership is needed for ethical ChatGPT: Character, assessment, and learning using artificial intelligence (AI)" is the third most-cited paper, with 188 citations. The article "The use of AI in education: Practicalities and ethical considerations" contains 61 citations, while "ChatGPT in medical imaging higher education" features 45 citations. Finally, "Use of ChatGPT in academia: Academic integrity hangs in the balance" has 44 citations (Table 5).

4. Risk of bias

We used the Cochrane risk of bias tool, Risk of Bias in Non-randomised Studies of Interventions (ROBINS-I), to assess bias in the included studies (Robinson et al., 2020). Other study designs, such as educational research, can also apply ROBINS-I to assess the quality and risk of bias (Schünemann et al., 2019).

ROBINS-I facilitates a thorough assessment of bias related to missing data, selective reporting, confounding, participant selection, categorisation, intervention, outcome measurement, reported results, and overall bias.

Each article was graded as "low", "some concerns", "high", or "no information". The traffic light system indicated the status of each article. Nested Knowledge, 2024 novel semi-automated software platform was used for the risk of bias analysis of this study.

A possible bias from different phases was analysed and presented in Fig. 3. The y-axis indicates the particular stage of the study process, while the x-axis measures each step's potential for bias using a colour-coded scale. The most significant indication of this process is the bias in participant selection. The study population may not accurately represent the population, or there may be an issue in the sample selection procedure, potentially due to factors such as limitations or the specific inclusion and exclusion criteria used to obtain the sample.

This study was conducted in the eight domains mentioned below:

D1: Bias due to confounding – There is a confounding effect of intervention in some studies. For example, the studies by Reiss (2021) and Verhoef (2022) may have bias due to confounding since the method of participant switching between the interventions was compared. Therefore, special care has been taken when interpreting the results.

D2: Bias in participant selection – The selection participants was based on participant characteristics and observed after the start of the intervention. There were nine studies in this category. Liu (2023) may have been biased when selecting participants, which may have affected the intervention and follow-ups with the participants. Maphoto et al. (2024) qualitative study had a large population of 14,000 and a sample size of 70. The sample size would have increased due to the large population. Other studies are shown in Fig. 4 in the yellow cycle and involve some concerns about participant selection which may have slightly impacted the outcomes.

D3: Bias in the classification of interventions – The information used to define intervention groups was recorded at the start of the intervention. The authors did not clearly define the intervention groups and may have missed the information recorded in the paper. Moreover, Reiss (2021) failed to specify how student groups engage with cameras. Marron (2023) was unable to provide more information about intervention groups with AI. Four studies fall under this category: Reiss (2021), Marron (2023), Crawford et al. (2023), and Verhoef et al. (2022).

D4: Bias in the measurement of outcomes – Deviations from the intended intervention beyond what would be expected in usual practice are considered. Liu et al. (2023) did not properly mention the intended intervention. Reiss (2021) divided to students with special educational needs.

There were seven studies in this domain: Liu et al. (2023), Reiss (2021), Avello-Sáez (2023), Marron (2023), Gamage (2023), Crawford et al. (2023), and Oravec (2022).

D5: Bias due to missing data – There were some missing data from the participants and intervention. This may have happened due to a lack of data availability or because participants were excluded from the analyses. Examples of studies with this bias are those by Liu et al. (2023), Reiss (2021), Marron (2023), Kelly et al. (2023), Bin-Nashwan et al. (2023), Birks and Clare (2023), Cotton et al. (2024), Oravec (2022). These studies are indicated by yellow circles in the traffic light system in Fig. 4. Furthermore, the participant information was not included by Weidmann (2024).

D6: Bias in the measurement of outcomes – The knowledge of the intervention could have influenced the outcome measure. The outcome measures may have affected participants' knowledge of the intervention, or the method of the outcome measurement may not have been applied equally among the participant groups. There are seven articles in this domain: Liu et al. (2023), Reiss (2021), Avello-Sáez (2023), Marron (2023), Gamage (2023), Crawford et al. (2023), Oravec (2022).

D7: Bias in the selection of the reported results – The reported effect estimates were likely selected based on the results from multiple outcome measurements within the outcome domain. The reported results may be biased in multiple outcome measures or the relationship between multiple outcome measures. Four studies were found in this domain: Liu et al (2023), Marron (2023), Birks (2023), and Verhoef (2022).

D8: Overall bias – Risk of bias judgement is considered in this domain. The following studies were included: Liu et al (2023), Alberth (2023), Marron (2023), Birks (2023), Crawford et al. (2023), Oravec (2022), and Verhoef (2022).

Table 5
Frequently cited papers.

Authors	Source Title	Citations
Cotton (2024)	Chatting and cheating: Ensuring academic integrity in the era of ChatGPT	755
Sullivan, M. (2023)	ChatGPT in higher education: Considerations for academic integrity and student learning	221
Crawford et al. (2023)	Leadership is needed for ethical ChatGPT: Character, assessment, and learning using AI	188
Reiss, Michael (2021)	The use of AI in education: Practicalities and ethical considerations	61
Currie, G (2023)	ChatGPT in medical imaging higher education.	46
Bin-Nashwan, S.A. (2023)	Use of ChatGPT in academia: Academic integrity hangs in the balance	44

5. Discussion

5.1. Population of the study

- *What characteristics do the participants have?*

This systematic review included 25 articles. Some of these were quantitative studies, and some were review papers. The total number of participants in this study was 2134, and 78 articles were considered for review articles. The population included medical students, radiologists, nuclear medicine technologists, scientists, students, professors in occupational therapy, postdoctoral researchers, fellows and scholars, lecturers and instructors, assistant professors, senior lecturers, and associate professors. The geographical areas of the research were Australia, New Zealand, the United States, the United Kingdom, South Africa, Nigeria, Namibia, Zimbabwe, India, the Congo, Ethiopia, and China (Table 6).

5.2. Intervention with AI tools

- *What specific AI tools and technologies are being implemented in academic activities?*

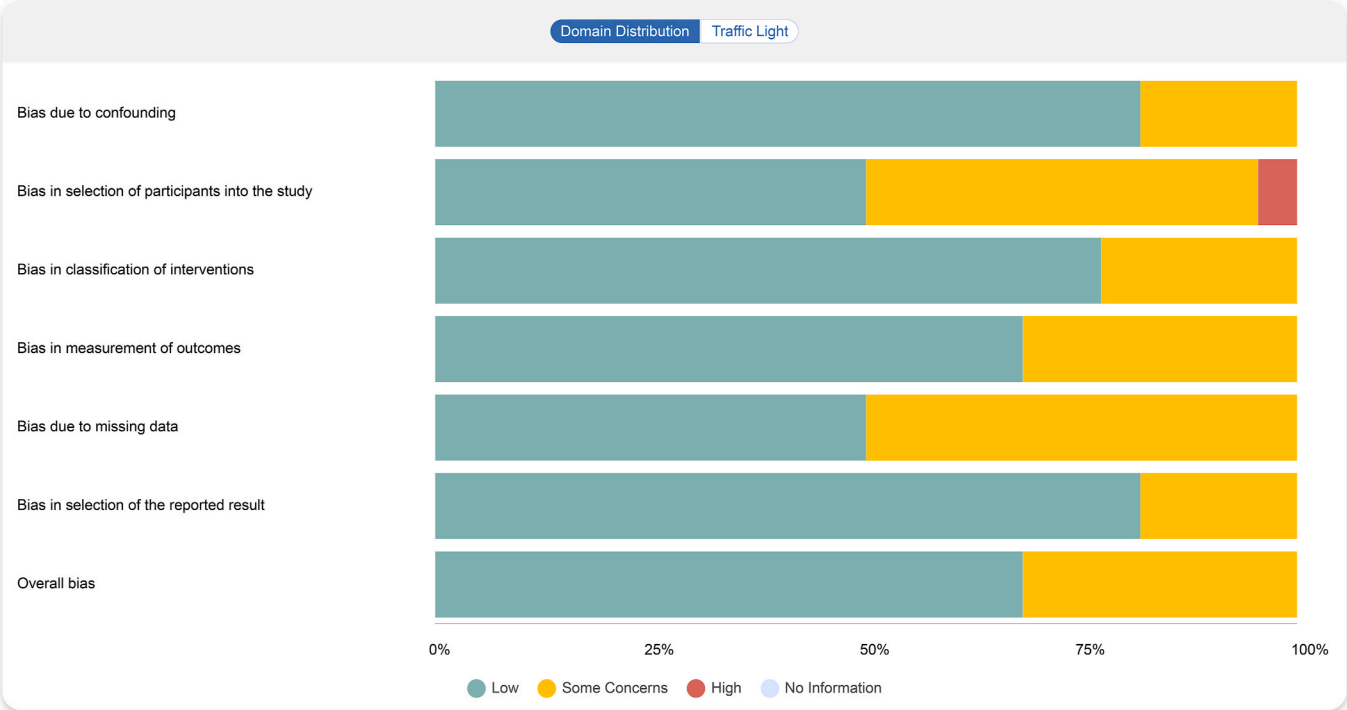

Fig. 3. Risk of bias domains' distribution.

Recently, generative AI, AI-powered chatbots, and writing tools have been gaining global attention (M. Liu et al., 2024a). Anyone, regardless of their designation, can use open-access AI tools if internet facilities are available (Avello-Sáez & Estrada-Palavecino, 2023), and since most people have internet access today, the barriers to using AI tools are minimal. These AI tools have been used for a variety of purposes (Choi et al., 2023), and several of them are used by students to perform academic tasks, including Writesonic, Jasper, Google Bard, Wordtune, Ryter, Writer, Anyword, Sudowrite (H. Liu et al., 2023a), ChatGPT (Sullivan et al., 2023; Avello-Sáez & Estrada-Palavecino, 2023; Salih, 2024), automated paraphrasing tools (Hammond et al., 2023), Microsoft's Bing Chat, Claude, Wit.ai, Hunyuan Aid, OPT by Meta, Alexa TM by Amazon, GPT-J and GPT-NeoX by EleutherAI, Megatron-Turing NLG by NVIDIA and Micros, LaMBDA (Gamage et al., 2023), Jenni AI, ResearchGPT, and Lateral (Weidmann, 2024).

Several websites provide automated paraphrasing tools (APTs). These websites encourage students by offering live academic writing demonstrations (Hammond et al., 2023). Students use these sites to save time and create more accurate writing. Specialised APT sites assist non-English-speaking students in completing their assignments, regardless of the subjects they are studying. Peer group encouragement, electronic word of mouth (e-WOM), and academic stress are significant factors in using AI tools for academic writing (Bin-Nashwan et al., 2023). New students in degree programmes may experience high levels of stress when adapting to a new environment (Barbayannis et al., 2022), which may motivate them to use AI tools to complete their assignments. Also, efficiency can be improved by using AI tools like ChatGPT (Kairu, 2022) since they save time, are easy to use, and utilise a large information pool (Salih, 2024). However, ChatGPT is not a solution for human critical thinking (Limna, 2023). Therefore, the use of AI tools raises serious issues concerning academic integrity.

The use of generative AI-powered writing tools significantly enhances academic writing by providing easily accessible and fast writing solutions for students worldwide.These tools provide sufficient support that saves time and reduces the academic stress placed on students (Maphoto et al., 2024). Therefore, students readily engage with these tools. However, the prominent use of AI tools in academic activities raises serious ethical concerns about academic integrity (Acosta-Enriquez et al., 2024), as these tools undermine the critical thinking skills of human beings and the quality of academic qualifications offered by institutes. Since the entire education sector navigates the technological landscape, efforts must be made to leverage AI tools' advantages while upholding the principles of academic integrity.

While students use AI tools to write, academics must use AI tools to detect AI writing and plagiarism. Most higher education institutes use Turnitin for this purpose (Currie et al., 2023; Verhoef et al., 2022).

5.3. Comparison of modern and traditional methods

What are the differences in academic integrity outcomes between assessments using AI and traditional methods?

In the 1950s, researchers started to investigate the methods and functions of AI (M. Liu et al., 2024b). However, Open AI's AI-powered chatbot (ChatGPT), released in November 2022, has had a profound impact on the education sector. Higher engagement was observed for ChatGPT (Sun & Hoelscher, 2023; M. Mijwil et al., 2023). As a result of the sudden boom of AI applications in the education sector, it became difficult to find students who do not use AI-related applications for academic work. Therefore, it is worth comparing AI applications for modern classroom assessment methods with traditional classroom assessment methods.

In traditional classrooms, teachers and students interact directly. As a result, teachers can monitor students' behaviour and detect academic misconduct. Many academic institutions have long-standing traditional codes of ethics and have built a strong academic integrity culture that students must become familiar with and respect. When students are in a modern learning environment, it is difficult to detect academic behaviour (Liu et al., 2023a; Bergström & Wiklund-Engblom, 2022) However, non-native English-speaking students showed a high tendency to breach academic integrity because they had difficulty writing their assignments in English (Sullivan et al., 2023; Herbold et al., 2023) These students used ChatGPT or similar software to write their assignments (Liu et al.,

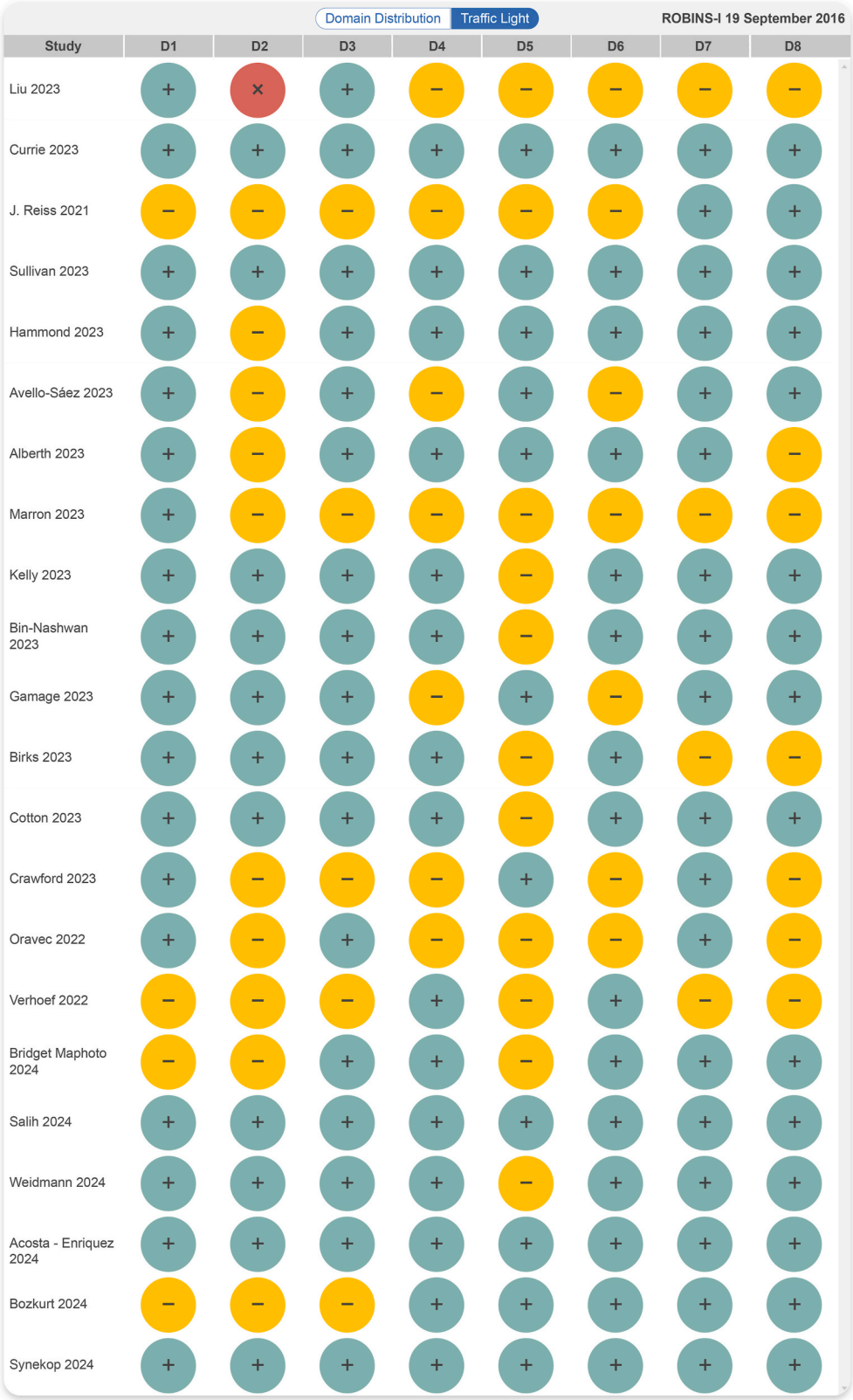

Fig. 4. Risk of bias traffic light overview.

2023). Meanwhile, academic institutions use Turnitin to detect AI writing in students' assignments (Table 6; Verhoef et al., 2022). Plug-ins allow for the direct integration of Turnitin into learning management systems, which makes it easy to install and use Turnitin with these systems.

Traditional exams are conducted in a classroom under controlled conditions. As a result, invigilators can directly monitor students' behaviour. In online exams, an online proctoring system must be used to monitor students' behaviour (Magalhães et al., 2020), and cameras are used to identify student behaviour in novel AI-based proctoring systems

Table 6
AI and academic integrity review summary.

No	First Author	Study Design	Participant Characteristics	Interventions/Comparators	Outcomes
1	Liu et al. (2023)	Review Study	Medical students	Interventions: AI: Writesonic, Jasper, Google Bard Wordtune, AI and chatbots, Comparator. The history of AI dates back to the 1950s.	Detection/prevention of cheating, ethical considerations in assessments, the proactive adoption of large language models
2	Currie (2023)	Qualitative study: science undergraduate course for both exams (n = 6) and written assignment tasks (n = 3).	N = 133 students (radiographers, nuclear medicine, technologists/ scientists)	AI augmentation of writing, risk of misconduct, academic integrity now and historically, AI challenged the traditional methods of learning and assessment. AI tools' use in the traditional model, Turnitin for similarity and AI scores, ChatGPT	Ethical considerations in assessments, ChatGPT poses a risk to academic integrity and the detection/prevention of cheating.
3	Reiss, Michael (2021)	Review study	No information	Cameras used to categorise students' moments, models of learning in the 19th century are outdated, AI	The possibility for AI to make the educational offer more precisely to a student's needs, AI provides the teaching, computer-based learning, discussed danger that AI education, narrow set of skills come to dominate
4	Sullivan, M. (2023)	Qualitative study: the article uses content analysis to examine news articles	Geography: Australia, New Zealand, the United States, and the United Kingdom; Sample size: N = 100	ChatGPT and non-traditional students in non-native English-speaking Australian schools have returned to pen-and-paper examinations.	Clear guidelines will need to be established, plagiarism stories in the media frequently occur, and universities need to be described as "updating", "reviewing", and "considering" their policies. Software tools can detect AI-written text.
5	Hammond et al. (2023)	Qualitative study	Web tools: QuillBot.com, plagiarismdetector.net, paraphrase-online.com,duplichecker.com, wordtune.com, N = 5	Large language model, generative AI tools, APT	Reconsider the definition of plagiarism concerning AI-generated text, empower students by fostering critical language awareness, and emotional appeals assist with attracting the user to examine and apply the tools to their writing needs.
6	Avello-Sáez, D. (2023)	Review study	Students and professors; school, occupational therapy	ChatGPT	Promoting academic integrity as part of the personal expectations of students, the attitudes and behaviours of classmates can highly influence a student's decision to cheat.
7	Alberth (2023)	Review study	No information	Paper writing, ChatGPT	Expectations to see these policies in place and international academic laws.
8	Marron (2023)	Review study	No information	ChatGPT, construct new knowledge, constructivism learning theory, and ChatGPT's benefits and challenges	Institutions must have balance to prevent academic misconduct and promote academic freedom and innovation, building a culture of academic integrity.
9	Kelly et al. (2023)	Mixed method	Australian university (N = 1135)	International students rarely used AI more than domestic students in science and engineering disciplines, degree of quality, Genai tools	AI should be used ethically, and student learning and academic integrity should be properly supported. Continuous monitoring, evaluations, and adaptations of teaching and assessment practices are necessary.
10	Bin-Nashwan et al. (2023)	Quantitative study	Postdoctoral researchers, fellows, and scholars (N = 702)	e-WOM, GPT-3	Social influence negatively affected academicians' use of ChatGPT, and e-WOM has a positive impact on ChatGPT usage in academia.
11	Gamage (2023)	Review study	N = 64 research articles	Microsoft's Bing Chat, OPT by Meta, Wit.ai, Hunyuan Aid, Alexa TM by Amazon, Claude, Bard GPT-J and GPT-NeoX by EleutherAI, Megatron-Turing NLG by NVIDIA and Micros, LaMBDA, and ChatGPT	A pre-established set of guidelines is needed. There is a need to re-design assessments and evaluation methods, check the patterns in the language or words, look for sources and citations, examine for uniqueness and novelty, check for language errors, and study the accuracy of AI content detective.
12	Birks and Clare (2023)	Review study	No information	ChatGPT	Policies and guidance are needed, the next generation of practitioners is needed, and universities need to store all text-based assessments submitted.
13	Cotton et al. (2024)	Review study	No information	GPT-3, inequities in the assessment process.	There is a need to check grammar, spelling and factual errors, check for originality, check for sources and citations, look for patterns or irregularities in the language, monitor student work closely, set clear guidelines for using AI, and use plagiarism detection tools.

(*continued on next page*)

Table 6 (*continued*)

No	First Author	Study Design	Participant Characteristics	Interventions/Comparators	Outcomes
14	Crawford et al. (2023)	Review study	No information	ChatGPT	AI does not substitute learning. There is increased pressure on students to succeed academically
15	Oravec, J.A. (2022)	Review study	US and UK.	AI, emerging cheating detection systems, traditional forms of dealing with cheating	AI should be used with cheating-detection systems. There is a need to develop homegrown systems of online cheating detection and use technological anti-cheating systems.
16	Verhoef, A.H. (2022)	Review study	South Africa	Online assessments, Turnitin, learning management system	There is a need to train students to avoid academic misconduct, promote a culture of academic integrity, and create discussion forums.
17	Maphoto et al. (2024)	Quantitative study	132 countries, including Nigeria, Namibia, Zimbabwe, India, Congo, Ethiopia, the US, and China (n = 70)	AI, ChatGPT, traditional and technological approaches to academic writing	Rethinking academic authenticity, creativity
18	Salih, M. (2024)	Quantitative study	Saudi faculty of medicine (n = 45)	AI applications	Cultural diversity and privacy/ confidentiality issues improve the learning experience
19	Weidmann (2024)	Review study	No information	AI-assisted writing	ethical considerations, academic integrity, authorship and the need for new legal frameworks
20	Acosta-Enriquez et al. (2024)	Quantitative study	University students (n = 201)	ChatGPT	There is a need to develop the competencies necessary to uphold academic integrity amid technological progress and to establish explicit policies and guidelines.
21	Bozkurt (2024)	Review study	No information	AI tools, traditional "publish or perish" culture development with AI	There is a need to use more than one method, such as oral exams and traditional writing assessments,
22	Synekop et al. (2024)	Quantitative study	ESP teachers (n = 22) and students (n = 60) in Ukraine	ChatGPT	Use plagiarism checkers, lack of awareness of academic integrity issues, and develop university policy regarding the use of AI models.
23	Herbold et al. (2023)	Quantitative study	Grammer school teachers (n = 139)	Chat GPT .4, Human writing and AI writing	Teachers need to adapt strategies for teaching writing skills
24	Yeadon et al. (2023)	Quantitative study	Department of Physics at Durham University. (n = 10)	ChatGPT and Davinci-003	Universities look for plagiarism rather than whether text was AI generated. Plagiarism detection tools cannot be relied upon AI generated work
25	Tan (2024)	Review study	No Information	AI tools	AI empower students to be motivated, ethical, and engaged learners.

(Reiss, 2021). Further, web browser access is limited, and the cut-and-paste option is disabled during examinations (Chirumamilla & Sindre, 2021). Additionally, most proctoring software has plug-ins for popular learning management systems like Moodle and Canvas. Conducting an online proctoring exam is not difficult in the modern virtual learning environment.

Traditional educational assessments are based on physical presence and proctoring, while the modern education system utilises technology-based solutions. However, academic misconduct can still occur in both systems. In modern education, if the institute does not use proper technology, it can be difficult to ensure students' identities and detect cheating.

5.4. Outcomes of AI use for academic activities

- *What ethical considerations arise from the integration of AI in academic assessments, and how are they addressed and mitigated?*

In academia, the most significant issue is the use of AI for writing. When students submit AI-written assignments and receive a degree, it can seriously impact the academic quality of the degree offered by the institution (Table 6; Currie, 2023). Therefore, it is important to discuss the preventive actions an academic institute can take to encourage students' academic integrity.

Academic institutes use Turnitin to detect academic misconduct. Turnitin reports indicate a similarity rate and AI content rate, though a low AI content rate may be a false positive. Yeadon et al. (2023) mentioned that plagiarism detection tools fail to detect AI-generated content properly. Therefore, lecturers should check the assignment content carefully. Turnitin offers support to detect academic misconduct, but language errors, words, citations, uniformity, and the novelty of the assignment must be checked (Table 6; Gamage et al., 2023; Cotton et al., 2024).

Academic cheating and detection have become thriving industries. Some companies develop AI writing and content humanising software, while others develop detection software. Therefore, it is not necessary to always use software to detect academic misconduct, as institutes can develop their own detection methods (Oravec, 2022). For example, an institute could develop manual procedures and train lecture panels to detect academic misconduct. Multiple methods, including oral exams and traditional writing tests like essays, can be used to detect AI academic misconduct (Bozkurt, 2024).

Institutes must clearly define what constitutes academic misconduct, including plagiarism, AI writing, and cheating, and they must establish clear policies and expectations (Abd-Elaal et al., 2019; Birks & Clare, 2023; Synekop et al., 2024). Then, the institute can implement an honour code with the agreement of students and faculty (Weidmann, 2024). Moreover, both teachers and students must agree to develop a policy for using AI on campus (Synekop et al., 2024). This can be officially documented and distributed among students and lecturers.

This is a good time to redefine academic integrity (Maphoto et al., 2024). The old definition does not match the new technology-based education. The creation of a culture of academic integrity is important in institutes (Verhoef et al., 2022; Rasu et al., 2023). In orientation

programmes, institutes should discuss academic integrity and introduce the policies to new students. Institutes also need to arrange regular workshops and training sessions for students and faculty about proper citation practices, academic ethics, plagiarism, and AI detection tools.

Since students are using AI for their academic work, there may be a need to revert to the old method of assignment checking to protect academic integrity. Institutions in Australia started implementing traditional examinations again to mitigate this issue (Cassidy, 2023). Moreover, ChatGPT is banned by some educational institutions (Tlili et al., 2023). It is difficult to prevent students from using ChatGPT in education (Sharples, 2022).

Therefore, the possibility of tailoring AI to students' needs must be acknowledged (Reiss, 2021). Liu et al. (2023) mentioned that using AI tools can improve medical students' knowledge. Meanwhile, ChatGPT improves the English writing of students (M. Liu et al., 2024a). The proper adaptation of the large language model has led to significant improvements in education. Therefore, academics must teach students how to use AI tools ethically (García-Peñalvo, 2023; Maphoto et al., 2024; Tan, 2024).

6. Future research

Although this paper discussed AI and academic integrity in the higher educational sector, future research areas can be identified in the same discipline. For instance, it is important to research new academic policies for the use of AI in educational settings. These can include the best practices and policies across different regions and cultural settings.

Future research can be conducted on how educational institutions should handle the intellectual property rights of AI-generated work. Such research will help to revise intellectual property rights and redefine authorship in the 21st century.

It is worth researching how students and faculty can ethically use AI to gain a deep understanding of the consequences of academic misconduct (Ali et al., 2024). AI-powered plagiarism detection tools like Turnitin can be used to detect complex forms of plagiarism, such as AI writing, paraphrasing, and translation plagiarism. Future research can be conducted on developing advanced algorithms that can better identify the semantics and context of the text, rather than merely matching content with internet sources.

It is also worth exploring the perceptions and attitudes of students and faculty regarding AI applications in education, as new training programmes can be designed based on the findings. Faculty members may express concerns about AI impact on academic integrity and make more human interaction during the training program to convince the students about the ethical use of AI (Zawacki-Richter et al., 2019).

Since new AI writing software tools with advanced AI-based algorithms are introduced every day, continuous research is required to investigate how new technologies detect AI-based cheating.

7. Limitations

This research has a few limitations. For one, the data collection process was limited to four databases, meaning relevant studies might have been missed. Moreover, AI software applications are rapidly changing as technology advances. As a result, the findings of this research may be outdated when new AI tools are introduced. Furthermore, the studies included in this research were conducted in Australia, New Zealand, the United States, the United Kingdom, South Africa, Nigeria, Namibia, Zimbabwe, India, Congo, Ethiopia, and China, which could harm the generalisability of the results to other countries, educational settings, and academic integrity definitions.

8. Conclusion

The integration of AI into education has positive and negative aspects. If students do not sufficiently support academics, it will be very difficult to uplift the academic integrity of institutes. AI tools such as ChatGPT certainly enhance the efficiency of writing and the accuracy of academic content, but they also violate academic integrity. Moreover, integrating AI into academic activities has the potential to improve students' learning abilities (Darvishi et al., 2024), and AI can help students improve their writing skills (Maphoto et al., 2024).

Furthermore, ChatGPT can be used to instruct students (Milano et al., 2023). Proper conversation encourages high student engagement. While some students may hesitate to ask their teachers questions, they may be willing to ask questions to ChatGPT without hesitation (Urban et al., 2024), potentially reducing their academic stress levels (Saif et al., 2024). Therefore, the use of ChatGPT or similar AI tools as virtual tutors is feasible. However, because of their limitations, AI tools cannot replicate human teachers.

AI has much potential to evolve (Bodily & Verbert, 2017), and its influence on education is unlimited. Therefore, educational institutes must create a culture of academic integrity to discourage academic misconduct and maximise the benefits of AI use in education. At the same time, traditional values such as honesty and trust must adapt to the modern technology-based environment to minimise academic misconduct. Even if an institute invests in proper AI detection software, traditional academic misconduct detection methods should still be used to improve academic integrity.

Finally, we believe that the ethical application of AI in academia is a collaborative effort of students, teachers, policymakers, and institutes. By promoting the ethical use of AI, institutions can uphold academic integrity and ensure that the qualifications they offer reflect students' knowledge and skills while enhancing the credibility of higher education in the digital era.

CRediT authorship contribution statement

Himendra Balalle: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sachini Pannilage:** Writing – original draft, Visualization, Data curation.

Data availability

Data will be made available on request.

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